

Federico — Hospital Administrative Operations Platform

Software Requirements Specification (SRS)

Team: 16_Federico

Version: 2.0 (Production-Grade Architecture & Multi-Tenancy Specification)

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1. INTRODUCTION

1.1 Purpose & Problem Statement

Hospitals face severe operational bottlenecks due to fragmented administrative workflows. Appointment booking, patient admission, bed allocation, non-clinical supply tracking, and billing typically operate in isolated silos without synchronized data flows. This leads to resource underutilization, long waiting periods, billing opacity, and high error rates. Federico addresses these challenges by establishing a single, unified administrative hub that orchestrates patient check-ins, real-time bed allocation, inventory management, dynamic role-based access control, and transparent billing workflows from pre-registration to final discharge.

1.2 Project Scope & Non-Clinical Focus

Explicit Scope Limitation: Federico is strictly a *non-clinical hospital operations system*. It manages administrative workflows, resource allocation, and billing settlement. It deliberately excludes all clinical diagnosis, medical prescriptions, laboratory analysis, and medical treatment planning. All clinical decision-making is handled by external Electronic Medical Record (EMR) systems.

1.3 Definitions and Acronyms

Term / Acronym	Definition
HMS	Hospital Management System.
SaaS	Software as a Service (Multi-Tenant Delivery Model).
UHID	Unique Health Identifier assigned to every registered patient.
PRE	Patient Registration & Eligibility Executive (Front-desk operations).
HOM	Hospital Operations Manager (Ward, bed, and service charge management).
FA	Finance Associate (Billing ledgers, insurance calculation, payment receipts).
DAL	Data Access Layer (Repository pattern abstraction over persistent storage).
RBAC	Role-Based Access Control with dynamic granular permission grants.

2. OVERALL SYSTEM DESCRIPTION

2.1 Multi-Tenant SaaS Architecture

Federico delivers multi-tenant capability supporting multiple hospital organizations simultaneously. Tenant context is resolved on every HTTP request via Bearer token session claims. Strict fail-closed tenant scoping ensures complete data isolation across organizations, while subscription tiers (Starter, Growth, Enterprise) dynamically enable modular feature flags (PHARMACY, LAB, INSURANCE, EMERGENCY).

2.2 User Personas & Role Matrix (6 Tiers)

Role	Scope	Primary Responsibilities
Platform Super User	Global SaaS	Tenant provisioning, plan management, module flags, global audit logs.
Hospital Admin	Tenant Owner	Hospital branding, branch configuration, custom dynamic RBAC, staff roles.
Hospital Operations (HOM)	Hospital Operations	Ward registry, real-time bed allocations, service charge logs, discharge sign-off.
Patient Registration (PRE)	Front Desk	Pre-registration review, OPD appointment scheduling, final discharge release.
Finance Associate (FA)	Financial Operations	Service charge review/approval, billing ledgers, insurance copays, receipts.
Patient	Self-Service	Appointment booking, admission requests, insurance uploads, online payments.

3. SYSTEM FEATURES & FUNCTIONAL REQUIREMENTS

3.1 Patient Registration, Intake & Scheduling

- **FR-1.1 (UHID Generation):** Every registered patient receives a unique, unalterable UHID.

- **FR-1.2 (Pre-Registration Triage):** Patients submit admission pre-requests (PENDING) reviewed by PRE staff (APPROVED/REJECTED).
- **FR-1.3 (OPD Scheduling):** PRE staff and patients schedule consultations against doctor availability slots with foreign key validation.

3.2 Real-Time Ward & Bed Management Cascade

- **FR-2.1 (Bed Allocation Cascade):** Allocating an AVAILABLE bed transitions bed to OCCUPIED, links pre-request to ADMITTED, creates admission record, and opens a financial ledger atomically.
- **FR-2.2 (Compensating Rollback):** If any step in the bed allocation cascade fails, the system executes a compensating transaction reverting bed status back to AVAILABLE.
- **FR-2.3 (Auto Bed Release):** Final PRE administrative discharge sign-off automatically frees the physical bed back to AVAILABLE.

3.3 Dynamic Role-Based Access Control (RBAC)

- **FR-3.1 (Custom Role Provisioning):** Hospital Admins can create tenant-specific custom roles with granular read/write/delete permission grants.
- **FR-3.2 (Zero Backdoor Access):** All API routes enforce dynamic RBAC grants and standard Bearer JWT claims; deprecated x-role headers are rejected.

3.4 Clinical Service Charge Logging & Finance Approval

- **FR-4.1 (Dual-Step Verification):** HOM logs service usage (Leader charges) which remain in PENDING state until FA explicitly reviews and approves them.
- **FR-4.2 (Ledger Insertion):** Approval atomically inserts the charge into the patient's billing ledger entries and updates running totals.

3.5 High-Performance Composite Endpoints & Client Resilience

- **FR-5.1 (Composite Summary API):** GET /patient/portal/summary returns profile, appointments, admissions, itemized bills, and receipts in a single roundtrip.
- **FR-5.2 (Double-Submit Protection):** Frontend API client enforces withAsyncLock to prevent duplicate form submissions and concurrent charges.
- **FR-5.3 (Request Timeout):** All network requests enforce 15-second AbortSignal timeouts.

4. NON-FUNCTIONAL REQUIREMENTS

Dimension	Requirement Specification
Performance	Composite portal endpoints respond under 50ms; debounced atomic writes prevent I/O blocking.
Security	OWASP Top 10 compliance, Bearer token authentication, input sanitization, fail-closed multi-tenancy.
Durability	ACID atomic disk persistence using staging temporary files (.tmp) and atomic filesystem swaps.
Usability	Reactive cross-tab state synchronization via storage events; bookmarkable URL hash routing (#/ledger/id).
Compatibility	Fully functional across Chrome, Microsoft Edge, and Firefox without native desktop plugins.

5. DATA ACCESS LAYER (DAL) & STATE MACHINES

The Data Access Layer encapsulates 12 specialized repositories inheriting from BaseRepository: PatientRepository, DoctorRepository, WardRepository, AppointmentRepository, PreRequestRepository, AdmissionRepository, BillingRepository, InventoryRepository, RbacRepository, OrganizationRepository, UserRepository, and ActivityLogRepository. All mutations trigger debounced crash-safe persistence.

6. VERIFICATION & AUTOMATED TESTING

Federico includes 17 automated Jest test suites encompassing 90 unit and integration tests with a 100% pass rate:

- **Test Coverage:** Data integrity, relational foreign key validation, multi-step transaction rollback, security middleware, and end-to-end multi-role patient lifecycle.
- **Automated Test Result:** 17 / 17 Test Suites Passed | 90 / 90 Tests Passed (100% Pass Rate).